

■ Churn Analysis (Churn Customers)

```
CREATE VIEW churn_customers AS
SELECT
    rc.customer_id,
    c.customer_full_name,
    c.mobile,
    c.age,
    c.assigned_staff_id,
    DATE(MAX(i.invoice_date))
    AS last_visited_date,
    DATEDIFF( (SELECT MAX(invoice_date) FROM invoices),
    MAX(i.invoice_date))
    AS Number_of_days_from_last_visit
FROM repeated_customers as rc
INNER JOIN customers_details as c
ON rc.customer_id=c.customer_id
INNER JOIN invoices as i
ON rc.customer_id=i.customer_id
GROUP BY rc.customer_id
HAVING Number_of_days_from_last_visit > 180
ORDER BY Number_of_days_from_last_visit DESC;
```

```
CREATE VIEW non_repeated_churn_customers AS
WITH customers AS (
    SELECT DISTINCT customer_id
    FROM customers_details
),
customer_details AS (
    SELECT c.customer_id, cc.age, cc.mobile,
    cc.customer_type, cc.referred_by_id,
    cc.assigned_staff_id
    FROM customers as c
    INNER JOIN customers_details as cc
    ON c.customer_id=cc.customer_id
),
customers_with_repeated_customers AS (
    SELECT cd.customer_id, rc.customer_id as repeated_customers,
    cd.age, cd.mobile,
    cd.customer_type, cd.referred_by_id,
    cd.assigned_staff_id
    FROM customer_details as cd
    LEFT JOIN repeated_customers as rc
    ON cd.customer_id=rc.customer_id
),
-- extracting_the_non_repeated_customers
extracting_the_non_repeated_customers AS (
    SELECT cwr.customer_id, cwr.repeated_customers,
    cwr.age, cwr.mobile,
    cwr.customer_type, cwr.referred_by_id,
    cwr.assigned_staff_id
    FROM customers_with_repeated_customers as cwr
```

```
FROM customers_with_repeated_customers as cwr
WHERE cwr.repeated_customers IS NULL
),
buyer_non_repeated_customer_details AS (
    SELECT enr.customer_id, enr.age, enr.mobile,
    enr.customer_type, enr.referred_by_id,
    enr.assigned_staff_id, i.invoice_id,
    i.invoice_date, i.total_amount_items,
    i.total_tax,
    i.final_bill_amount
    FROM extracting_the_non_repeated_customers as enr
    INNER JOIN invoices as i
    ON enr.customer_id=i.customer_id
)
SELECT
    bnrcd.customer_id, bnrcd.age, bnrcd.mobile,
    bnrcd.customer_type, bnrcd.referred_by_id,
    bnrcd.assigned_staff_id,
    DATE(MAX(bnrcd.invoice_date)) AS Last_visit_date,
    DATEDIFF( (SELECT MAX(invoice_date) FROM invoices),
    MAX(bnrcd.invoice_date))
    AS Number_of_days_from_last_visit
FROM buyer_non_repeated_customer_details as bnrcd
GROUP BY bnrcd.customer_id, bnrcd.age, bnrcd.mobile,
    bnrcd.customer_type, bnrcd.referred_by_id,
    bnrcd.assigned_staff_id
HAVING Number_of_days_from_last_visit > 180
ORDER BY Number_of_days_from_last_visit DESC;
```

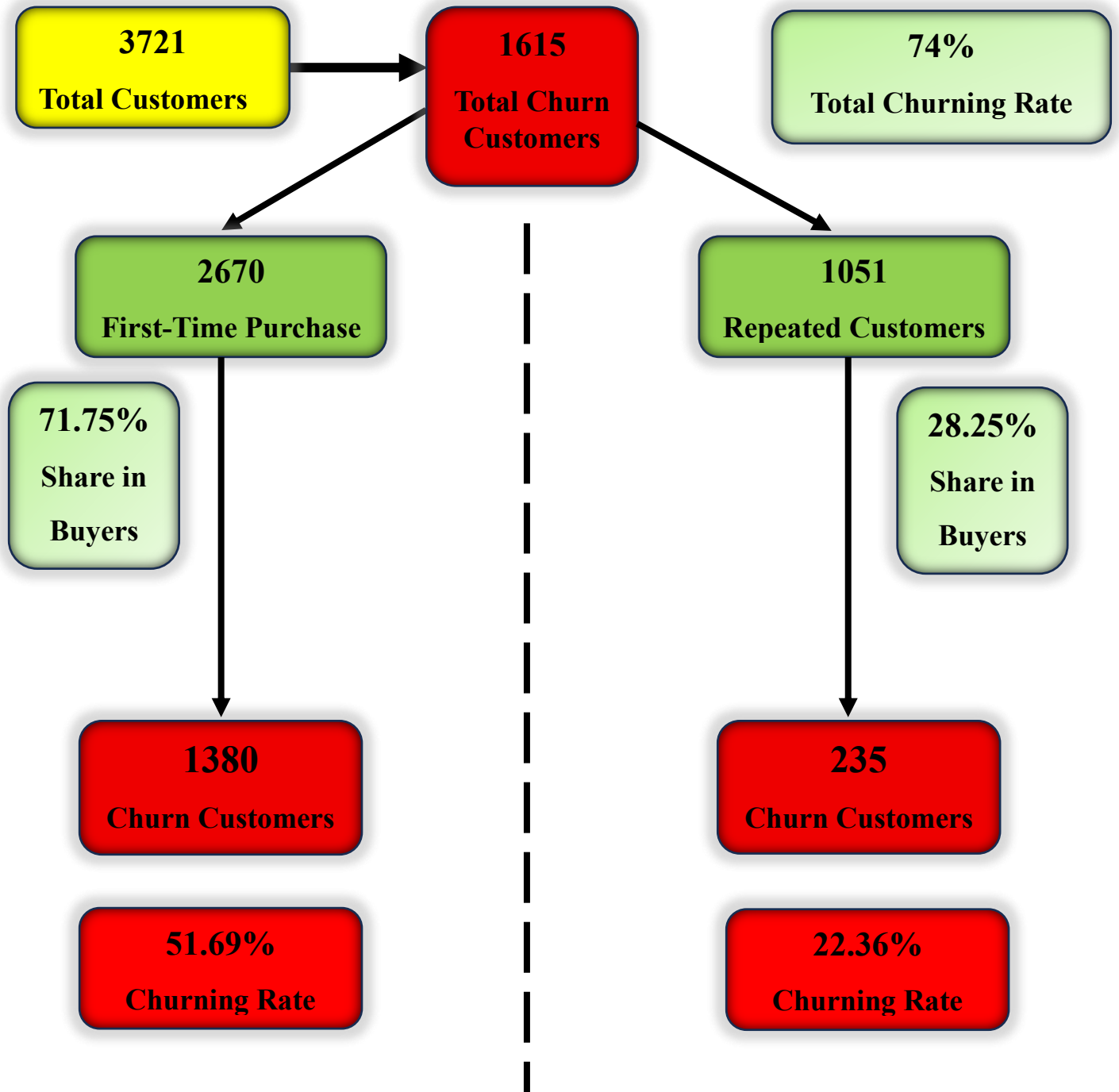
Before diving into the churn analysis queries, a dedicated VIEW named Churn customers was created to store all the churned customer records in a reusable virtual table. The VIEW identifies every repeated customer who has not made a purchase in more than 180 days classifying them as churned based on the difference between the store's most recent invoice date and each customer's last purchase date.

NOTE - Assumption: Since the dataset is historical, customer inactivity and churn are calculated relative to the latest transaction date in the dataset (2026-01-11) rather than the system's current date CURDATE(). This avoids overstating inactivity due to the absence of data beyond the dataset period.

Business Questions Answer about the Churning Customers are as follows:

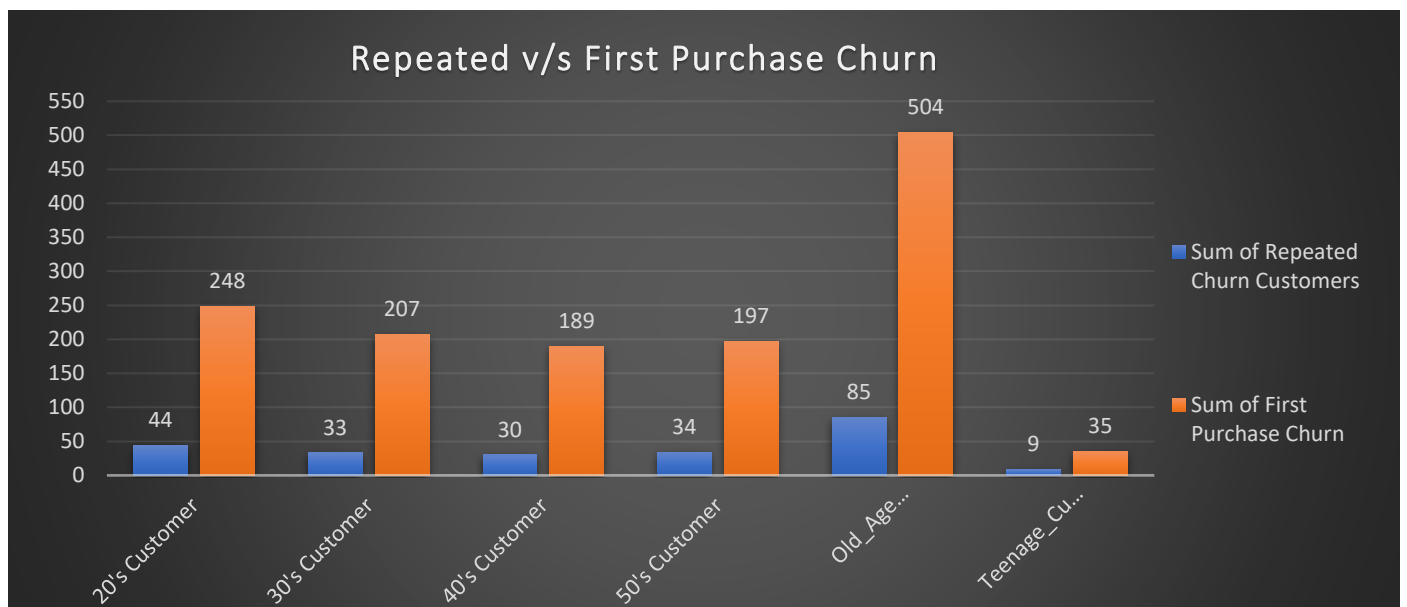
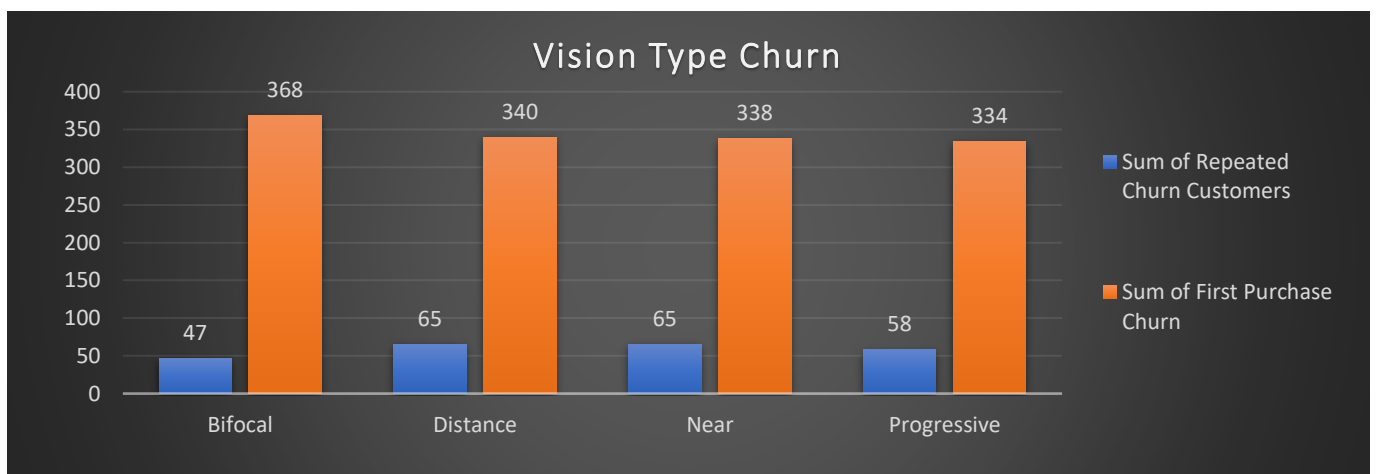
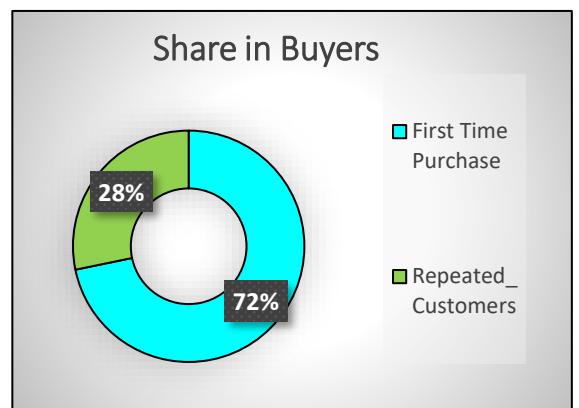
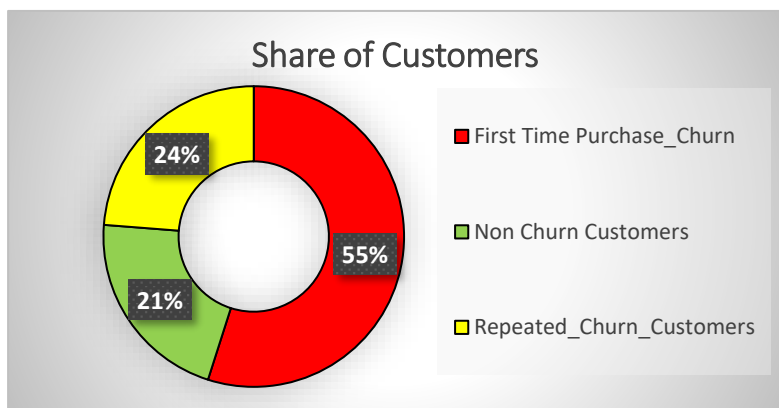
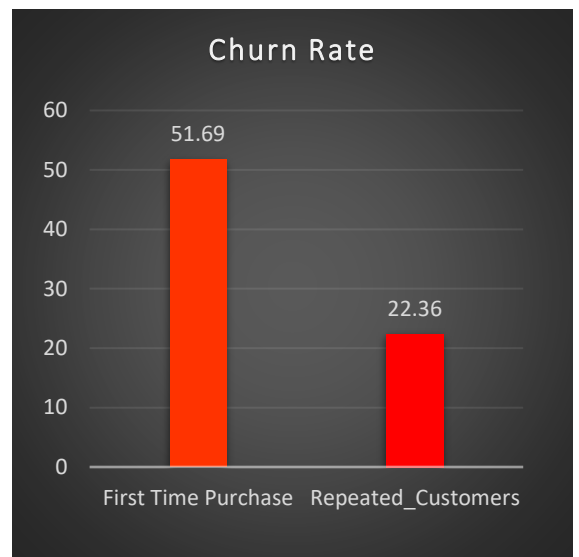
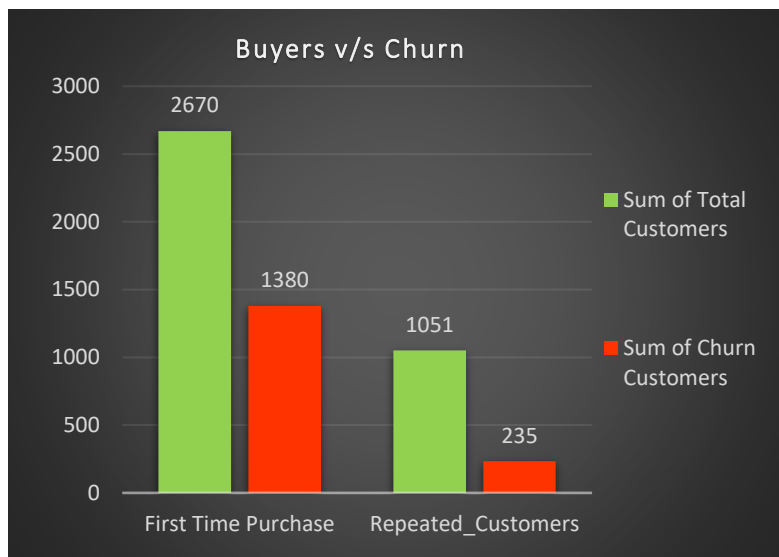
- 1) How many of our repeated customers those who have previously demonstrated loyalty by purchasing more than once have not returned in over 180 days and can now be classified as churned, & what percentage of our 1,051 repeated customers does this represent?
- 2) How much revenue did these churned customers generate before they stopped visiting and what is the total revenue the business is at risk of losing permanently?
- 3) Which age group has the highest number of churned customers and does a particular generation show a stronger tendency to disengage from the store over time?
- 4) Which churned customers were among our top revenue generators and who should be prioritized first for re-engagement outreach?
- 5) How many repeated customers are currently in the "At Risk" zone having not visited in 90 to 180 days and can still be recovered before they churn completely?
- 6) What is the average number of days since the last purchase across all churned customers and how long ago did the business begin losing these customers?
- 7) Which staff members have the highest number of churned customers under their assignment and does staff performance play a role in customer retention?
- 8) Which vision type is most commonly associated with churned customers and does the type of eye condition influence long term loyalty?

Dashboard of Churning Customers (Churn Analysis)

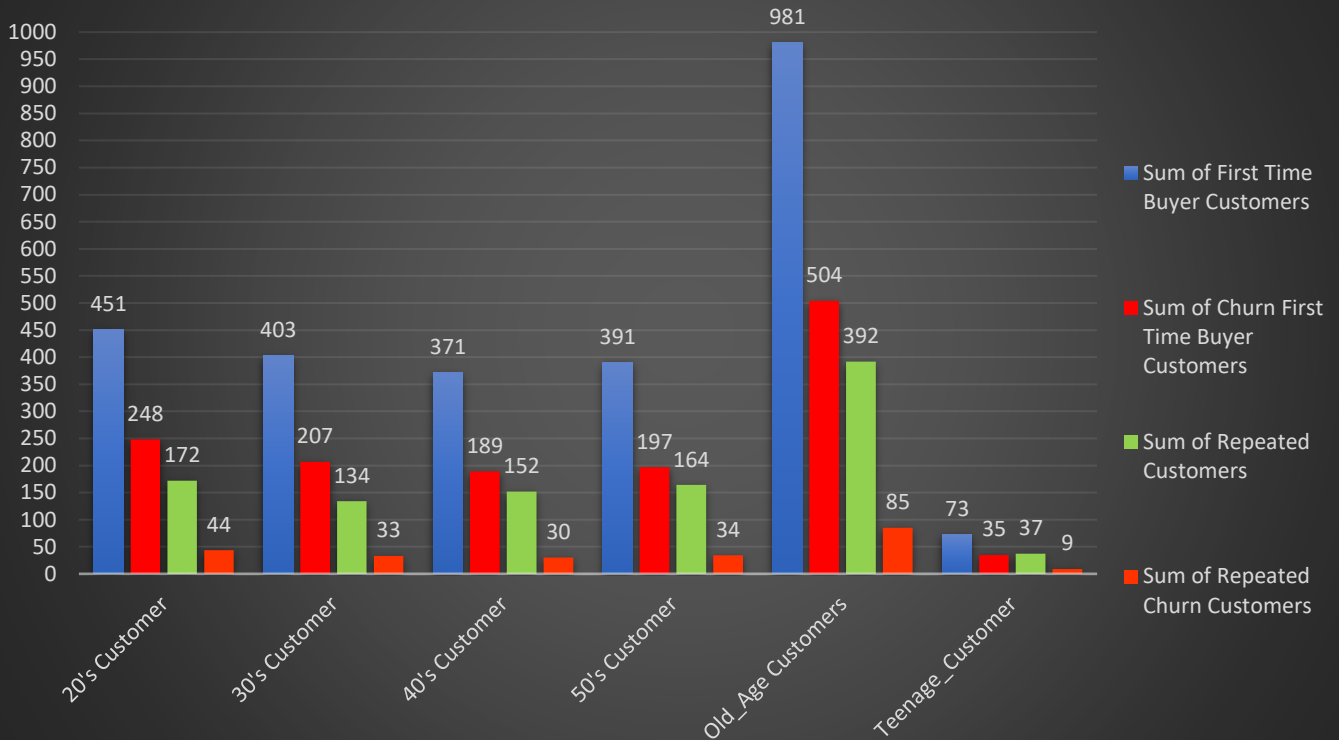


SQL QUERY OUTPUT:

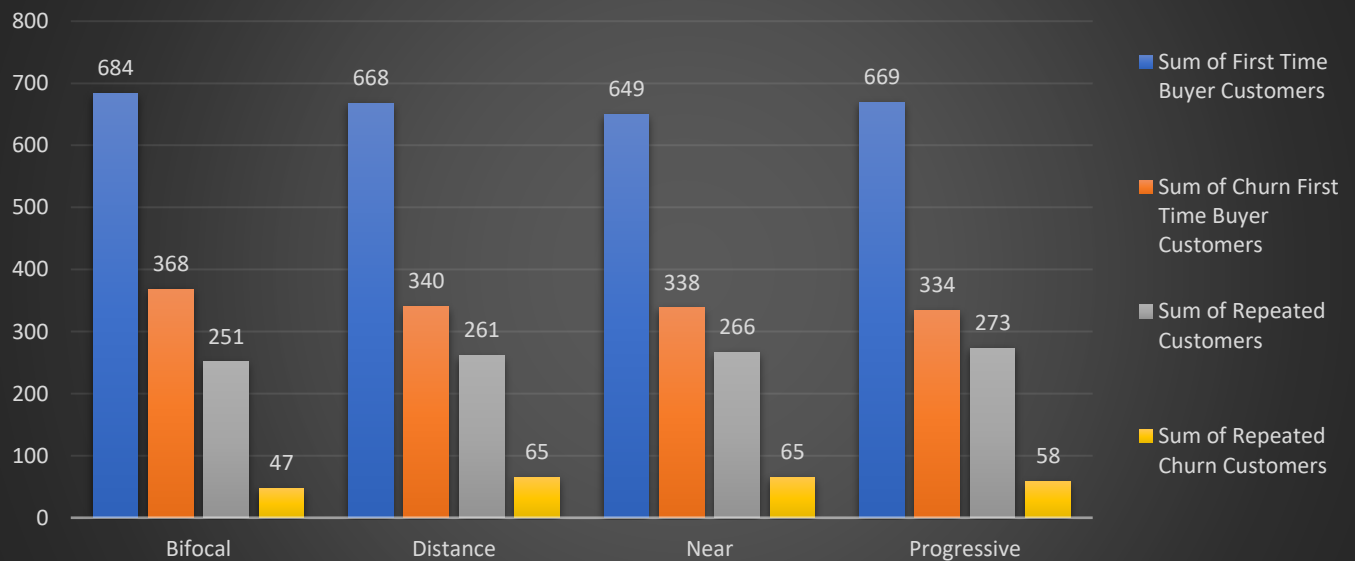
Result Grid Filter Rows: Export: Wrap Cell Content:					
	Churn_Customer_Type	Total_Customers	Share_in_Buyers	Churn_Customers	Churn_Rate
►	First Time Purchase	2670	71.75	1380	51.69
	Repeated_Customers	1051	28.25	235	22.36
	Total	3721	100.00	1615	74.05



Churn Customers across Age group (First time buyer v/s Repeated)



Churn Customers across Vision Type (First Time Buyers v/s Repeated Buyers)



Result Grid							
Filter Rows:		Export:		Wrap Cell Content:			
	Age_group	First_time_buyers	Churn_Customers	Churn_Rate	repeated_customers	Churn_Customers	Churn_Rate
	HULL	2670	1380	51.69	1051	235	22.36
	Old_Age Customers	981	504	51.38	392	85	21.68
	20's Customer	451	248	54.99	172	44	25.58
	30's Customer	403	207	51.36	134	33	24.63
	50's Customer	391	197	50.38	164	34	20.73
	40's Customer	371	189	50.94	152	30	19.74
	Teenage_Customer	73	35	47.95	37	9	24.32

Result Grid							
Filter Rows:		Export:		Wrap Cell Content:			
	vision_type	First_time_buyers	Churn_Customers	Churn_Rate	repeated_customers	Churn_Customers	Churn_Rate
	HULL	2670	1380	51.69	1051	235	22.36
	Bifocal	684	368	53.80	251	47	18.73
	Progressive	669	334	49.93	273	58	21.25
	Distance	668	340	50.90	261	65	24.90
	Near	649	338	52.08	266	65	24.44

Churn Customer Analysis

Why this Analysis Matters

Customer churn is one of the most critical business metrics because acquiring a new customer is significantly more expensive than retaining an existing one. In this project, repeat customers have already been identified as one of the store's most valuable customer segments, contributing nearly half of the total business revenue despite representing only a small proportion of buyers.

Therefore, understanding which customers are leaving the business, why they are leaving, and which customers are at risk of churning enables management to take proactive actions before valuable customer relationships are permanently lost.

The objective of this analysis is to identify churn patterns across customer demographics, vision types, revenue contribution, customer value, and staff performance so that targeted retention strategies can be implemented.

Key Insight 1 – Most Churn Comes from First-Time Buyers

Out of 3,721 buyer customers, the analysis identifies 1,615 churned customers, resulting in an overall churn rate of 43.4% (*don't use the summed 74.05%; use 1615/3721 instead*). Among these:

- 1,380 are First-Time Buyers.
- 235 are Repeated Customers.

This means approximately 85% of all churned customers are first-time buyers, while only 15% are repeated customers.

Business Insight:

The largest customer leakage occurs immediately after the customer's first purchase.

Many customers purchase once but never return, indicating that the business performs well in customer acquisition but has significant room for improvement in converting first-time buyers into loyal customers.

Strategic Recommendation

Management should strengthen the post-purchase customer journey by implementing:

- Follow-up calls or WhatsApp messages
- Customer satisfaction surveys
- Eye check-up reminder campaigns
- Personalized product recommendations
- Loyalty enrollment after the first purchase

The objective should be to convert more first-time buyers into repeat customers.

Key Insight 2 – Repeated Customers Churn Less but Their Loss is More Costly

Although repeated customers account for only 28.25% of total buyers (1,051 customers), only 235 of them have churned, resulting in a churn rate of 22.36%.

This churn rate is significantly lower than the 51.69% observed among first-time buyers.

However, repeated customer churn represents a much greater business risk.

Previous analysis showed that:

- Repeated customers represent only 22% of buyers
- Yet they generate 47% of total business revenue

Therefore, every repeated customer lost represents the loss of multiple future purchases rather than a single transaction.

Business Insight

Repeated customers demonstrate substantially higher loyalty than first-time buyers.

However, because of their high lifetime value, losing even a small proportion of repeated customers can significantly reduce long-term profitability.

Strategic Recommendation

Repeated customers should receive dedicated retention initiatives, including:

- Personalized follow-ups
- Annual eye examination reminders
- Prescription renewal campaigns
- VIP loyalty benefits
- Relationship-based customer service

Protecting this customer segment should remain one of the business's highest strategic priorities.

Key Insight 3 – Senior Customers Record the Highest Churn Volume Across Both Customer Types

When churn is analysed across age groups, Senior Customers (60+) consistently record the highest number of churned customers in both categories.

This finding aligns with previous demographic analysis, where senior customers also represented the largest customer base.

Business Insight

The higher churn volume among senior customers is primarily driven by their larger population rather than poor loyalty.

Despite having the highest number of churned customers, senior customers also remain:

- The largest revenue contributors
- The largest repeat customer segment
- The largest referral source

Strategic Recommendation

Rather than treating senior customer churn as a weakness, management should view it as the largest retention opportunity.

Improving retention within this already large customer base can produce significant long-term revenue growth.

Key Insight 4 – First-Time Buyer Churn is Consistently High Across Every Age Group

Across every age segment, first-time buyer churn is substantially higher than repeated customer churn.

Business Insight

This consistent gap indicates that the primary business challenge is not retaining loyal customers it is converting first-time buyers into repeat customers.

Once customers return for a second purchase, their likelihood of remaining engaged with the business increases considerably.

Strategic Recommendation

The business should redesign its first-time customer experience by focusing on:

- Better onboarding
- After-sales engagement
- Educational content
- Follow-up reminders
- Building trust immediately after the first purchase

Key Insight 5 – Churn is Evenly Distributed Across Vision Types

Churn analysis across vision types shows relatively similar customer counts for:

- Bifocal
- Distance
- Near Vision
- Progressive

No single vision category dominates customer churn.

This indicates that churn is not driven by a particular vision condition but is instead influenced by overall customer relationship management.

Business Insight

The business serves a well-diversified vision portfolio without excessive dependence on any single vision segment.

Key Insight 6 – Progressive Customers Require Special Retention Attention

Although Progressive Vision customers do not record the highest churn volume, they deserve the greatest strategic attention.

Previous analysis showed that Progressive lenses generate:

- The highest revenue
- Higher profit margins
- Greater customization requirements
- Strong alignment with senior customers

Because Progressive dispensing requires precise eye testing, accurate PD measurement, expert fitting, and skilled consultation, even one poor customer experience can permanently lose a high-value customer.

Strategic Recommendation

The business should continue investing in:

- Experienced optometrists
- Precision diagnostic equipment
- Staff training
- Accurate lens fitting
- Premium customer consultation

Protecting Progressive customers will preserve one of the store's highest-margin revenue streams.

Overall Business Conclusion

The churn analysis reveals two distinct business challenges:

First-Time Buyer Churn

- Represents the largest source of customer loss.
- Requires improvements in post-purchase engagement and customer conversion.
- Every additional repeat customer created today strengthens future revenue.

Repeated Customer Churn

- Occurs at a much lower rate.
- However, it carries disproportionately high financial impact because repeated customers generate 47% of total business revenue.
- Retaining these customers should remain one of the business's highest strategic priorities.

Executive Comparison (Highlight Box)

Critical Business Finding

- 71.75% of buyers are first-time purchasers, yet 51.69% of them churn after their initial purchase.
- 28.25% of buyers become repeated customers, and only 22.36% of them churn.
- Repeated customers contribute 47% of total business revenue, making them nearly twice as valuable as their population share suggests.

Management implication:

1. The business must pursue a dual strategy reduce first-time buyer churn to create more repeat customers, and aggressively retain existing repeat customers because they are the primary drivers of long-term profitability.
2. Protect Senior Customers, who remain the largest source of recurring revenue and referrals.
3. Strengthen Progressive Lens Customer Experience, where both revenue potential and service sensitivity are highest.

The analysis clearly shows that sustainable business growth will depend not only on acquiring new customers, but on successfully converting first-time buyers into repeat customers and protecting the high-value customers who already generate the majority of long-term revenue.

Additional Churn Analysis

Beyond demographic and vision-type analysis, additional SQL analyses were performed to identify customers requiring immediate management attention.

These analyses include:

1. Churned Repeat Customers

- Number of previously loyal customers who have now churned
 - Percentage of churn among the 1,051 repeat customers
-

2. Revenue at Risk

- Total historical revenue generated by churned customers
 - Estimated revenue permanently at risk if these customers are not recovered
-

3. High-Value Churn Customers

- Identification of churned customers who previously generated the highest revenue
 - Priority list for personalized re-engagement campaigns
-

4. Customers Currently "At Risk"

- Customers inactive between 90–180 days
- Customers who can still be recovered before becoming fully churned

This provides management with an early-warning system rather than waiting until customers are permanently lost.

5. Average Customer Inactivity

- Average number of days since the last purchase across all churned customers

This helps management understand how long customers remain inactive before the relationship is effectively lost.

6. Staff-wise Churn Analysis

- Number of churned customers assigned to each staff member

This analysis helps determine whether customer retention differences are associated with sales staff performance, follow-up quality, or relationship management practices.

Overall Business Conclusion

The churn analysis indicates that the business currently maintains a relatively healthy repeated customer base, with an overall churn rate of **22.36%**, which is comparatively very high, since repeated customers based is the one which has contributed 47% revenue of the business, hence losing this 22% of the customers is like something losing a very high portion of the revenue. However, the analysis highlights clear opportunities to improve long-term customer lifetime value through targeted retention initiatives.

The most important strategic priorities are:

1. **Protect Senior Customers**, as they generate the highest revenue, repeat purchases, referrals, and lifetime value.
2. **Closely monitor Progressive Lens customers**, where service quality directly influences customer retention and profitability.
3. **Implement proactive re-engagement campaigns** for customers in the 90–180 day "At Risk" window before they become permanently inactive.
4. **Prioritize recovery of high-value churned customers**, as recovering an existing profitable customer is significantly more cost-effective than acquiring a new one.
5. **Increase engagement with Teenage Customers** through fashion-oriented marketing and digital channels to reduce their relatively high churn rate.

Overall, the findings reinforce a key business principle identified throughout this project: sustainable growth depends not only on attracting new customers, but on retaining valuable existing customers, preventing churn, and maximizing customer lifetime value.

Additional Churn Analysis:

1) Extracting the Churn Customers details & % in repeated customers

1) How many of our repeated customers those who have previously demonstrated loyalty by purchasing more than once have not returned in over 180 days and can now be classified as churned, and what percentage of our 1,051 repeated customers does this represent?

```
WITH churning_customers AS (  
  SELECT  
    rc.customer_id,  
    DATE(MAX(i.invoice_date)) AS Last_Visit_Date,  
    DATEDIFF(CURDATE(), MAX(i.invoice_date))  
    AS Days_from_last_visit  
  FROM repeated_customers as rc  
  INNER JOIN invoices as i  
  ON rc.customer_id=i.customer_id  
  GROUP BY rc.customer_id  
  HAVING Days_from_last_visit > 180  
  ORDER BY Days_from_last_visit DESC  
)  
churning_customers_details AS (  
  SELECT  
    cc.customer_id, c.customer_full_name,  
    c.mobile, c.age, cc.Last_Visit_Date,  
    cc.Days_from_last_visit  
  FROM churning_customers as cc  
  INNER JOIN customers_details as c  
  ON cc.customer_id=c.customer_id  
)  
SELECT  
  ccd.customer_id, ccd.customer_full_name,  
  ccd.mobile, ccd.age, ccd.Last_Visit_Date,  
  ccd.Days_from_last_visit  
FROM churning_customers_details as ccd;
```

```
WITH churning_customers AS (  
  SELECT  
    rc.customer_id,  
    DATEDIFF( (SELECT MAX(invoice_date) FROM invoices ),  
    MAX(i.invoice_date)) AS Number_of_days_from_last_visits  
  FROM repeated_customers as rc  
  INNER JOIN invoices as i  
  ON rc.customer_id=i.customer_id  
  GROUP BY rc.customer_id  
  HAVING Number_of_days_from_last_visits > 180  
  ORDER BY Number_of_days_from_last_visits DESC  
)  
SELECT  
  COUNT( DISTINCT rcc.customer_id)  
  AS Total_Repeated_Customers,  
  COUNT( DISTINCT CASE WHEN cc.customer_id IS NOT NULL THEN cc.customer_id END)  
  AS Churning_repeated_customers,  
  ROUND( COUNT( DISTINCT CASE WHEN cc.customer_id IS NOT NULL THEN cc.customer_id END)* 100 /  
  COUNT( DISTINCT rcc.customer_id), 2) AS Percentage_of_Churn_Customers  
FROM repeated_customers as rcc  
LEFT JOIN churning_customers as cc  
ON cc.customer_id=rcc.customer_id;
```

customer_id	customer_full_name	mobile	age	Last_Visit_Date	Days_from_last_visit
7325	Saumya Karpe	0568512814	49	2025-01-26	520
3293	Vedant Saxena	5480227836	22	2025-02-06	509
4169	Hitesh Korpai	1624156656	42	2025-02-06	509
1166	Quincy Bala	7314923403	51	2025-02-07	508
3944	Eesha Banik	2033560086	38	2025-02-10	505
3986	Maanas Brar	1643364188	22	2025-02-11	504
6266	Mugdha Datta	1866781344	34	2025-02-13	502
1809	Leena Dugal	9621641944	76	2025-02-16	499
1111	Krish Bahl	0183544535	60	2025-02-19	496
6616	Kala Vora	2127629587	45	2025-02-19	496
1629	Niharika Nath	6059400213	75	2025-02-20	495
6708	Caleb Konda	9203467054	65	2025-02-22	493
7005	Chasmum Wali	0166012578	53	2025-02-22	493
2325	Anvi Tara	3461214400	30	2025-02-24	491
7329	Indrajit Devan	1076196510	75	2025-02-25	490
4664	Jagrati Narula	7546971956	51	2025-02-26	489
344	Kiaan Madan	5929300620	23	2025-02-27	488
2764	Chakravedy Das	3541300349	62	2025-02-27	488
4500	Dominic Kothari	1735730331	53	2025-02-27	488
1867	Madhav Palani	2414597827	30	2025-03-01	486
7522	Amruta Minhas	7940878896	33	2025-03-07	480

Total_Repeated_Customers	Churning_repeated_customers	Percentage_of_Churn_Customers
1051	235	22.36

2) Total revenue the business is at risk of losing permanently

2) How much revenue did these churned customers generate before they stopped visiting and what is the total revenue the business is at risk of losing permanently?

```
SELECT  
  SUM(i.final_bill_amount)  
  AS Revenue_from_Churn_Customers,  
  ROUND( SUM(i.final_bill_amount)*100/  
  (SELECT SUM(final_bill_amount) FROM invoices ), 2)  
  AS Percentage_of_revenue_at_lost  
FROM churn_customers as cc  
INNER JOIN invoices as i  
ON cc.customer_id=i.customer_id;
```

Revenue_from_Churn_Customers	Percentage_of_revenue_at_lost
4779880.00	10.21

3) Count of Churned Customers in each Age Group

3) Which age group has the highest number of churned customers and does a particular generation show a stronger tendency to disengage from the store over time?

```
SELECT
CASE
WHEN cc.age < 21 THEN "Teenage_Customers"
WHEN cc.age BETWEEN 21 AND 30 THEN "20's Customers"
WHEN cc.age BETWEEN 31 AND 40 THEN "30's Customers"
WHEN cc.age BETWEEN 41 AND 50 THEN "40's Customers"
WHEN cc.age BETWEEN 51 AND 60 THEN "50's Customers"
WHEN cc.age > 60 THEN "Old_Age_Customers"
END AS Age_Group,
COUNT(DISTINCT cc.customer_id) AS COUNT_Customers
FROM churn_customers as cc
GROUP BY Age_Group WITH ROLLUP
ORDER BY COUNT_Customers DESC;
```

Result Grid			 Filter Rows:	
	Age_Group		COUNT_Customers	
▶	NULL		235	
	Old_Age_Customers		85	
	20's Customers		36	
	50's Customers		34	
	30's Customers		33	
	40's Customers		30	
	Teenage_Customers		17	

4) Top Revenue generating Churn Customers

4) Which churned customers were among our top revenue generators and who should be prioritized first for re-engagement outreach?

```
SELECT
cc.customer_id,
cc.customer_full_name,
cc.mobile, cc.age,
SUM(i.final_bill_amount)
AS Revenue
FROM churn_customers as cc
INNER JOIN invoices as i
ON cc.customer_id=i.customer_id
GROUP BY cc.customer_id, cc.customer_full_name,
cc.mobile, cc.age
ORDER BY Revenue DESC;
```

Result Grid		Filter Rows:	Export:	Wrap
customer_id	customer_full_name	mobile	age	Revenue
3293	Vedant Saxena	5480227836	22	47956.00
2141	Tarak Garg	1670533401	23	42550.00
7111	Vritti Acharya	8450100202	35	42134.00
5487	Unni Borde	3865031997	82	41777.00
5465	Hemangini Doshi	4075937519	18	41068.00
982	Ayaan Mann	9018662089	69	39833.00
3702	Meghana Pall	5368327615	29	39563.00
478	Jhalak Tandon	5864182497	23	38887.00
4074	Isha Menon	8924518051	18	38518.00
4500	Dominic Kothari	1735730331	53	38430.00
5666	Omisha Keer	6298702456	50	38315.00
344	Kiaan Madan	5929300620	23	37826.00
2809	Orinder Sankaran	4379917915	55	37719.00

5) Customers "AT RISK Zone" of getting Churn (90 to 180 Days)

5) How many repeated customers are currently in the "At Risk" zone having not visited in 90 to 180 days and can still be recovered before they churn completely?

```
WITH at_risk_customers AS (
SELECT
i.customer_id,
DATEDIFF( (SELECT MAX(invoice_date) FROM invoices),
MAX(i.invoice_date) )
AS Number_of_days_from_last_visit
FROM invoices as i
GROUP BY i.customer_id
HAVING Number_of_days_from_last_visit BETWEEN 90 AND 180
ORDER BY Number_of_days_from_last_visit DESC
),
at_risk_customers_details AS (
SELECT
arc.customer_id, c.customer_full_name,
c.mobile, c.age,
arc.Number_of_days_from_last_visit
FROM customers_details as c
INNER JOIN at_risk_customers as arc
ON arc.customer_id=c.customer_id
)
SELECT arcd.customer_id, arcd.customer_full_name,
arcd.mobile, arcd.age,
arcd.Number_of_days_from_last_visit
FROM at_risk_customers_details as arcd;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
customer_id	customer_full_name	mobile	age	Number_of_days_from_last_visit
205	Leela Ghosh	4666327309	25	180
3494	Nikita Ravel	9273115352	76	180
4507	Christopher Prasad	6018976926	84	180
6824	Arunima Atwal	9524762839	31	180
7730	Girindra Mahal	0046592511	32	180
1237	Chandran Magar	2030964277	68	179
1777	Meera Kannan	8045864144	65	179
2393	Saksham Wadhwa	6016130067	41	179
2815	Triya Gulati	7459102014	29	179
4493	Advik Prasad	4558652297	72	179
7466	Caleb Barman	4955725308	66	179
479	Yashica Shah	6103731956	34	178
1833	Arunima Sampath	6223898537	41	178
2862	Oni Palan	1098790386	27	178
3287	Jonathan Raghavan	5385009733	52	178
4895	Jackson Kar	5021228343	83	178
4899	Darsh Mitra	5060164446	59	178
4935	Chandhal Jhaveri	7569313495	76	178

6) Average Days of all the Churn Customers from last Visit

6) What is the average number of days since the last purchase across all churned customers and how long ago did the business begin losing these customers?

```
SELECT
    COUNT(DISTINCT cc.customer_id)
    AS Total_Churned_Customers,
    AVG(cc.Number_of_days_from_last_visit)
    AS Average_days_since_last_purchase,
    MAX(cc.Number_of_days_from_last_visit)
    AS Longest_Absence,
    MIN(cc.Number_of_days_from_last_visit)
    AS Shortest_Absence,
    MIN(cc.last_visited_date)
    AS Earliest_Churn_date,
    MAX(cc.last_visited_date)
    AS Most_recent_Churn_date
FROM churn_customers as cc;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Total_Churned_Customers	Average_days_since_last_purchase	Longest_Absence	Shortest_Absence	Earliest_Churn_date	Most_recent_Churn_date
				235	239.6128	350	181	2025-01-26	2025-07-14

7) Staff with the Highest number of Churn Customers

7) Which staff members have the highest number of churned customers under their assignment and does staff performance play a role in customer retention?

```
WITH staff_revenue AS (
SELECT
    i.staff_id, s.staff_full_name,
    SUM(i.final_bill_amount) AS staff_revenue
FROM invoices as i
INNER JOIN staff as s
ON i.staff_id=s.staff_id
GROUP BY i.staff_id, s.staff_full_name
ORDER BY staff_revenue DESC
)
SELECT
    cc.assigned_staff_id, sr.staff_full_name,
    sr.staff_revenue,
    COUNT(DISTINCT cc.customer_id)
    AS COUNT_Churn_Customers
FROM churn_customers as cc
INNER JOIN staff_revenue as sr
ON sr.staff_id=cc.assigned_staff_id
GROUP BY cc.assigned_staff_id, sr.staff_full_name,
sr.staff_revenue
ORDER BY COUNT_Churn_Customers DESC;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	assigned_staff_id	staff_full_name	staff_revenue	COUNT_Churn_Customers
				12	Naksh Gandhi	2151305.00	16
				16	Pratyush Kata	2401868.00	15
				19	Anita Vala	2146680.00	15
				1	Neel Varty	2398491.00	14
				5	Dhriti Mohan	2478569.00	14
				20	Utkarsh Hayre	2500938.00	14
				3	Luke Kata	2411549.00	13
				9	Neha Lalla	2197589.00	13
				13	Indali Keer	2323801.00	13
				2	Anjali Amble	2114982.00	12
				4	Vinaya Lalla	2343254.00	12
				15	Netra Lal	2471046.00	12
				14	Gautam Choud...	2682251.00	11
				17	Anmol Gole	2549710.00	11
				6	Sai Keer	2373953.00	10
				8	Mohini Srinivasan	2288597.00	10
				11	Tara Swamy	2479416.00	9
				7	Robert Sawhney	2325696.00	8
				10	Neelima Sibal	1836366.00	7
				18	Owen Narayanan	2335352.00	6

8) Count of Churn Customers based on Vision Type

8) Which vision type is most commonly associated with churned customers and does the type of eye condition influence long term loyalty?

```
SELECT
    p.vision_type,
    COUNT(DISTINCT cc.customer_id)
    AS Churn_Customers
FROM churn_customers as cc
INNER JOIN prescriptions as p
ON cc.customer_id=p.customer_id
GROUP BY p.vision_type
ORDER BY Churn_Customers DESC;
```

Result Grid			Filter Rows:
	vision_type	Churn_Customers	
▶	Near	65	
	Distance	65	
	Progressive	58	
	Bifocal	47	